

## ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

## Chapter 2: Non-Directional Beacon (NDB) & ADF Navigation

Syllabus Source: CAE Oxford (pp. 45-88) | Pass Target:  $\geq 90\%$  AviationExam

### 1. NDB GROUND STATION & ADF AIRBORNE RECEIVER

The NDB is an omnidirectional transmitter operating in the LF/MF bands (190 to 1750 kHz). The airborne receiver is the Automatic Direction Finder (ADF):

- **Antenna Principle:** Uses a combined Loop Antenna (figure-8 reception pattern with two nulls) and Sense Antenna (circular pattern). Combining them produces a CARDIOID (heart-shaped) pattern with a single unique null, eliminating  $180^\circ$  ambiguity.
- **Relative Bearing (RB):** The angle measured clockwise from the aircraft's nose to the station.
- **Magnetic Bearing TO Station (QDM):**  $QDM = \text{Magnetic Heading (MH)} + \text{Relative Bearing (RB)}$   
(If sum  $> 360^\circ$ , subtract  $360^\circ$ !).
- **Magnetic Bearing FROM Station (QDR):**  $QDR = QDM \pm 180^\circ$ .

### 2. ADF OPERATIONAL ERRORS & LIMITATIONS

- **Night Effect (The Biggest Hazard):** At night (especially dawn/dusk), sky waves reflected by the ionosphere interfere with the ground wave. The ADF needle hunts, oscillates violently, or points  $30^\circ$  to  $90^\circ$  off course!
- **Coastal Refraction:** Radio waves travel faster over water than over land. When crossing a coastline obliquely, the wave bends TOWARDS THE COASTLINE NORMAL. The needle indicates the station is closer to the coast than it actually is!
- **Thunderstorm / Static Error:** Lightning discharges emit massive LF/MF radio noise. The ADF needle swings away from the NDB and POINTS DIRECTLY TOWARDS THE CUMULONIMBUS CLOUD!
- **Mountain Effect & Quadrantal Error:** Mountain reflections create false nulls. Quadrantal error is caused by aircraft metal bending the incoming wave (zero at nose, tail, and wingtips; maximum on quadrants:  $045^\circ$ ,  $135^\circ$ ,  $225^\circ$ ,  $315^\circ$ ).

### 3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

#### [\*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating QDM and Intercepting NDB Inbound Radial

SCENARIO (Classic AviationExam Navigation Problem): An aircraft is flying on Magnetic Heading (MH) =  $040^\circ$ : - The Relative Bearing Indicator (RBI) needle points to  $220^\circ$  ATC instructs: 'Intercept and track inbound on the  $070^\circ$  QDM to the NDB with a  $30^\circ$  intercept angle.' QUESTION 1: What is the current QDM and QDR? QUESTION 2: What heading should the pilot turn to intercept the  $070^\circ$  QDM at  $30^\circ$ ? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate current QDM (Bearing TO the station): - Formula:  $QDM = \text{Magnetic Heading (MH)} + \text{Relative Bearing (RB)}$  -  $QDM = 040^\circ + 220^\circ = 260^\circ$ ! - ( $QDR = 260^\circ - 180^\circ = 080^\circ$  from the station). Step 2: Understand where you are relative to target QDM ( $070^\circ$ ): - Current QDM =  $260^\circ$  (You are Southwest of the station, bearing  $260^\circ$  to it). - Target QDM =  $070^\circ$  (You need to fly inbound towards  $070^\circ$ ). Step 3: Calculate Intercept Heading (Inbound track  $070^\circ$  with  $30^\circ$  intercept): - Target Track Inbound =  $070^\circ$ . - You are on the right/south side of the desired track. - Intercept Heading = Inbound Track -  $30^\circ = 070^\circ - 30^\circ = 040^\circ$  Magnetic! FINAL ANSWER: Current QDM =  $260^\circ$ . Intercept heading =  $040^\circ$  Magnetic.

**[!] EXAM TRAP (AVIATIONEXAM) - QDM vs QDR Memory Trap**

- QDM = Bearing TO the station (Inbound track to fly in zero wind). - QDR = Bearing FROM the station (Radial outbound from station). - Always check if the question asks for QDM or QDR: QDR = QDM +/- 180°!

**4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**

**Q1: An aircraft is flying on a magnetic heading of 120°. The ADF needle points to relative bearing 080°. What is**

[A] 040° [B] 200° (QDM = MH + RB = 120° + 080° = 200°) [C] 020° [D] 300°

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. Formula: QDM = MH + RB = 120° + 80° = 200° magnetic bearing to the station.

**Q2: Why does the ADF suffer from severe 'NIGHT EFFECT' errors at dawn and dusk?**

[A] Cold air bends the wave [B] SKY WAVES reflected by the ionosphere interfere with the ground wave, causing needle hunting and wild bearing errors [C] Aircraft generators produce static [D] Moisture in the antenna

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. At night and twilight, ionospheric reflection permits sky waves to reach the receiver with shifting polarization, degrading the loop antenna null.

**Q3: When flying in the vicinity of active thunderstorm activity, what does an ADF needle tend to do?**

[A] Points to Magnetic North [B] POINTS TOWARDS THE LIGHTNING ELECTRICAL DISCHARGES in the Cumulonimbus cloud [C] Spins continuously at 100 rpm [D] Freezes at 000°

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. Lightning creates powerful electromagnetic pulses in the LF/MF band, causing the ADF sense loop to track the storm cell rather than the NDB.